environmental microbiology reports

Environmental Microbiology Reports (2022) 14(3), 376–384 doi:10.1111/1758-2229.13016

Brief Report

Genomic diversities of *ctx*B, *tcp*A and *rst*R alleles of *Vibrio cholerae* O139 strains isolated from Odisha, India

Dipti Ranjan Behera, Ashish Kumar Nayak, Smruti Ranjan Nayak, Dilena Nayak, Sipraswati Swain, Pradeep Kumar Maharana, Bhagyalaxmi Biswal, Swatishree Pany, Sanghamitra Pati and Bibhuti Bhusan Pal*
Microbiology Division, ICMR-Regional Medical Research Centre, Chandrasekharpur, Bhubaneswar, OR, 751023, India.

Summary

The genome of *Vibrio cholerae* O139 strains has undergone cryptic changes since its first emergence in 1992 in South India. This study aimed to determine the presence of genotypic changes marked in *ctx*B, *tcp*A and *rst*R genes located within the CTX prophages among the strains of *V. cholerae* O139 isolated from 1999 to 2017 in Odisha. Antibiotic susceptibility test was conducted on 59 *V. cholerae* O139 strains. A conventional PCR assay was done for *ctx*B gene typing followed by sequencing along with identification of *rst*R and *tcp*A gene. Pulsed-field gel electrophoresis (PFGE) was carried out to reveal clonal variations among the *V. cholerae* O139 strains. Among *V. cholerae* O139 isolates more than 60% showed resistance to ampicillin, co-trimoxazole, furazolidone, streptomycin, neomycin and nalidixic acid. The *ctx*B sequencing and *rst*R allele-specific PCR assay revealed the presence of three genotypes 1, 3 and 4 with at least one copy of CTX Calc φ in addition to CTX ET and CTX Cl prophages in *V. cholerae* O139 isolates. PFGE analysis revealed 13 pulsotypes with two clades having 60% similarity among *V. cholerae* O139 strains. The circulating *V. cholerae* O139 strains in Odisha showed variation in genotypes with multiple clonal expansions over the years.

Received 30 July, 2021; accepted 5 October, 2021. *For correspondence. E-mail bbpal_rmrc@yahoo.co.in; Tel. (+91) 674 2305638; Fax (+91) 674 2301351.

Introduction

Cholera, an acute dehydrating diarrhoea caused by the aquatic bacterial pathogen *Vibrio cholerae*, dates back to antiquity in the form of epidemics and pandemics throughout the World (Politzer, 1959; Kaper *et al*., 1995; Faruque *et al*., 1998a). *Vibrio cholerae* has been classified into more than 200 serogroups based on the somatic O antigen from which O1 and O139 serogroups produce cholera toxin and thus carries pathogenic strains (Yamai *et al*., 1997). In the beginning of 19th century, the *V. cholerae* O1 El Tor biotype was supposed to be responsible for all epidemics and endemic cholera worldwide. Later, the *V. cholerae* O139 appeared during 1992 in South India by replacing the existing *V. cholerae* O1 strains, rapidly spread to various cholera prone regions in India, Bangladesh and neighbouring countries (Ramamurthy *et al*., 1993; Nair *et al*., 1994). The *V. cholerae* O1 and O139 serogroups are indistinguishable specifically in the cholera toxin (CT) and toxin co-regulated pili (TCP) encoded by the CTX prophage (Faruque *et al*., 1994; Faruque *et al*., 2000). The occurrence of two new genotypes of *V. cholerae* O139 from 1993 to 2005 was first documented in Kolkata, where isolates of *V. cholerae* O139 carried Calcutta type (CTX^{Calc}φ) CTX phage (genotype 4) and a new hybrid genotype 5 due to a novel combinations of *ctx*B and *rst*R alleles (Raychoudhuri *et al*., 2010) (Fig. 1). Later on the genotypic changes in cholera toxin were reported from Delhi that showed majority of *V. cholerae* O139 strains harboured CT genotype 1 and 3 with a new clone of genotype 4 (*rst*R^{Calc}) (Ghosh *et al*., 2016). Similar findings from Guangdong, China also revealed the presence of a novel genotype 5 of *V. cholerae* O139 serogroup rather than genotype 1 and 3 during 1993–2013 (Li *et al*., 2016). However, a new genotype of *V. cholerae* O139 carrying mutation in the prototype El Tor *ctx*B gene was reported during cholera outbreak in 2017 from Odisha (Pal *et al*., 2019). So, continuous changes in the genotypic characteristics of *V. cholerae* O139 have been reported across the South-Asian cholera endemic countries based on the *ctx*B and *rst*R allelic patterns that determine the CTXφ of this serogroup. The present study has been envisaged to study the genomic diversities of *ctx*B, *rst*R alleles

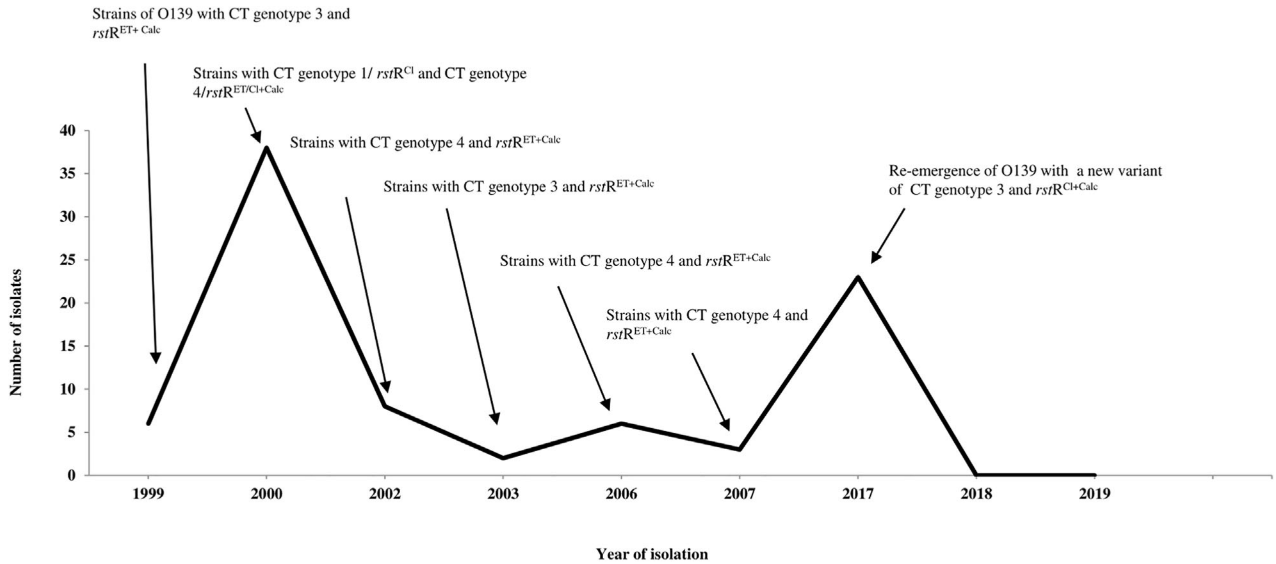


Fig. 1. Prevalence of *rst*R alleles and CT genotypes among *Vibrio cholerae* O139 strains isolated from clinical and environmental sources.

along with *tcp*A genes present in the *V. cholerae* O139 isolates from 1999 to 2017 in Odisha, which will offer new insights into the biology of *V. cholerae* O139 strains.

Results

Antibiotic susceptibility

The antibiotic susceptibility assay on the *V. cholerae* O139 strains showed that more than 60% of the isolates were resistant to ampicillin, furazolidone, neomycin and nalidixic acid. The *V. cholerae* O139 isolates of 2002 and 2006 differed in their antibiotic susceptibility patterns from the rest of the study period by displaying resistance to co-trimoxazole, furazolidone, nalidixic acid and neomycin only. However, in the subsequent years (2007 and 2017), all the isolates exhibited resistance additionally to ampicillin and streptomycin (Table 1).

Molecular characterization

The mPCR-I revealed that the *V. cholerae* O139 strains isolated from 1999 to 2017 were positive for *rtx*C, *rfb*O139, *ctx*B, *omp*U and *tox*R genes, whereas mPCR-II showed the presence of *omp*W, *ctx*A, *tcp*A and *zot* genes with some variations. The *rfb*O139 gene confirmed the O139 serogroup in all the isolated strains, whereas *omp*W gene is specific for *V. cholerae* at species level. The PCR based study on the distribution of virulence genes *in V. cholerae* O139 strains over the years revealed some variations, including *rtx*C (95.2%), *omp*U (95.2%), *tox*R (99.3%) and *tcp*A (95.9%). Further, the analyses of antibiotic-resistant genes *Sul*II, *dfr*A1, *SXT* and *Str*B present in the *V. cholerae* O139 strains showed little variations. The *Str*B gene, which provides resistance for streptomycin, was 91.1% positive, while SXT was 81% positive. The *dfr*A1 gene was present in 60.4% strains, whereas *Sul*II gene was positive in 88.4% strains that provided resistance towards trimethoprim and sulfamethoxazole respectively (Table 1).

Distribution of ctxB genotypes

The mismatch amplification mutation assay (MAMA) PCR assay on all the *V. cholerae* O139 strains showed an increasing trend of El Tor specific *ctx*B3 allelic patterns in *V. cholerae* O139 isolates in 2017. In contrast, only 16.9% of *V. cholerae* O139 strains showed classical *ctx*B1 allelic patterns as noticed in 2000 and 2006, while 35.6% of the O139 strains yielded amplicons for both the allelic types *ctx*B1 and *ctx*B3 in 2000, 2002, 2006 and 2007. The *V. cholerae* O139 strains isolated in 2017 yielded amplicons only with the El Tor-specific *ctx*B primers (Table 2).

Analysis of tcp*A gene*

The PCR analyses of *tcp*A gene specific for *V. cholerae* O139 isolates yielded a 167 bp amplicon with their respective separate forward primers. El Tor-specific *tcp*A showed dominance from 1999 to 2017 (62.7%) except in 2003 that showed classical *tcp*A. The first appearance of Haitian *tcp*A in *V. cholerae* O139 strains was noticed in 2000 and 2006 only (8.5%). The classical *tcp*A in *V. cholerae* O139 showed its presence only in 2003 and 2017 (28.8%) (Table 2).

Table 1. Antibiotic resistance profiles and genotypic characteristics of *Vibrio cholerae* O139 isolates (1999–2017).

			Virulent genes									Antibiotic-resistant genes			
			mPCR I					mPCR II							
Year of isolation	No. of strains (*n*)	Antibiogram profile	*rtxC*	*rfbO139*	*ctxB*	*ompU*	*toxR*	*ompW*	*ctxA*	*tcpA*	*zot*	*SulII*	*dfrA1*	*SXT*	*StrB*
1999	3	AmFrSN	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)
2000	21	CFrSNCoNA	+ (95%)	+ (100%)	+ (90%)	+ (95%)	+ (95%)	+ (71%)	+ (95%)	+ (100%)	+ (100%)	+ (95%)	+ (76%)	+ (100%)	+ (95%)
2002	4	FrNNA	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)
2003	2	AmFrSCoNA	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)
2006	5	CoNA	+ (100%)	+ (80%)	+ (80%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)
2007	3	AmFrSNA	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)	+ (100%)
2017	21 (14 + 7)[a]	AmFrSNNA	+ (71.4%)	+ (71.4%)	+ (100%)	+ (71.4%)	+ (100%)	+ (100%)	+ (71.4%)	+ (71.4%)	+ (71.4%)	+ (24%)	+ (47%)	+ (67%)	+ (43%)

[a] Out of 21 *Vibrio cholerae* O139 strains, seven strains were isolated from environmental water bodies.
Am, ampicillin; C, chloramphenicol; Co, co-trimoxazole; NA, nalidixic acid; Fr, furazolidone; S, streptomycin; N, neomycin.

Analysis of rstR alleles

The repressor gene (*rst*R) carried by RS2 flanking region of the 7.0-kb long circular CTXφ genome is essential for the regulation of CTXφ. The diversification of nucleotide sequences of *rst*R gene revealed three different alleles, such as CTX^{Cl}, CTX^{ET} and CTX^{Calc} type. The analysis of *rst*R gene revealed that *V. cholerae* O139 isolates from 1999 to 2017 carried at least one copy of Calcutta type CTX^{Calc} prophage in addition to the CTX^{ET} and CTX^{Cl} prophages. However, 57.6% of the *V. cholerae* O139 isolates yielded amplicons for both the *rst*R^{ET} and *rst*R^{Calc} types over the study periods. Interestingly, 18.6% of the O139 strains isolated in 2000 and 2017 possessed both alleles *rst*R^{Cl} and *rst*R^{Calc} types, whereas two isolates from 2000 and nine from 2017 also yielded mixed amplicons for all three *rst*R allele types, i.e., *rst*R^{Cl}, *rst*R^{ET} and *rst*R^{Calc} (Table 2).

Nucleotide sequence analysis of ctxB gene

The nucleotide sequencing of the *ctx*B gene of 10 representatives *V. cholerae* O139 strains (Table 3) from the above-mentioned periods showed that one O139 isolate each from 1999, 2000, 2003 and 2017 produced an amplicon for El Tor-specific primers of *ctx*B. These isolates were designated as genotype 3, possessing nucleotides A, T, T at positions 83, 115 and 203 that correspond to amino acids, such as aspartic acid, tyrosine and isoleucine at positions 28, 39 and 68 respectively. One *V. cholerae* O139 strain isolated in 2000 yielded amplicons for classical *ctx*B (genotype 1) with nucleotides A, C, and C at positions 83, 115 and 203 that corresponds to amino acids, such as aspartic acid, histidine and threonine at positions 28, 39 and 68, respectively. Five *V. cholerae* O139 strains from 2000 to 2007 (two strains in 2000, one strain each from 2002, 2006 and 2007) produced amplicons for both classical and El Tor *ctx*B (genotype 4) due to overlapping sequence peaks of C/A, C/T and C/T at nucleotide positions 83, 115 and 203, that showed alteration in amino acids, such as alanine, histidine and threonine at positions 28, 39 and 68, respectively. The amino acid sequence alignments that showed changes at positions 28, 39 and 68 for the respective genotypes 1, 3 and 4 is depicted in Fig. 2.

PFGE analysis

The pulsed-field gel electrophoresis (PFGE) patterns of the representative *V. cholerae* O139 isolates from 1999 to 2017 exhibited two distinctly different clusters with 13 pulsotypes with 60% similarity. Among two clades, one consisted of *V. cholerae* O139 strains from 1999 to

Table 2. Distribution of different types of *ctxB*, *rstR* and *tcpA* genes among *Vibrio cholerae* O139 strains from Odisha: 1999–2017.

		Types of *ctxB* genes			Types of *rstR* genes						Types of *tcpA* genes		
Year of isolation	No. of strains	Only El Tor type	Only Classical type	Both Classical and El Tor type	Only El Tor type	Only Classical type	Only Calcutta type	El Tor, Classical and Calcutta type	Both El Tor and Calcutta type	Both Classical and Calcutta type	El Tor type	Classical type	Haitian type
1999	3	3	0	0	0	0	0	0	3	0	3	0	0
2000	21	2	8	11	1	2	0	2	12	4	18	0	3
2002	4	0	0	4	0	0	0	0	4	0	4	0	0
2003	2	2	0	0	0	0	0	0	2	0	0	2	0
2006	5	0	2	3	0	0	0	0	5	0	3	0	2
2007	3	0	0	3	0	0	0	0	3	0	3	0	0
2017	21	21	0	0	0	0	0	9	5	7	6	15	0
Total (%)	59	28 (47.4%)	10 (16.9%)	21 (35.6%)	1 (1.7%)	2 (3.4%)	0	11 (18.6%)	34 (57.6%)	11 (18.6%)	37 (62.7%)	17 (28.8%)	5 (8.5%)

2006 (cluster A), whereas the other cluster consisted of strains from 2017 only (cluster B) (Fig. 3).

Discussion

The emergence of non-O1 *V. cholerae* that caused sporadic cholera cases across the South-Asian region with multiple clonal expansions formed the impetus to study the divergent genotypic characteristics of *V. cholerae* O139 strains isolated from Odisha. The number of *V. cholerae* O139 strains remained very less over the years except in 2000 and 2017 which contributed to diarrheal outbreaks in coastal as well as tribal areas of Odisha. Analysis of the *V. cholerae* O139 isolates from 1999 to 2017 showed changes in the antimicrobial resistance patterns throughout the years. The strains isolated in 1999 were resistant to ampicillin but changes occurred in successive years and again resistance to ampicillin reappeared with a gap of 4 years in 2003. Similarly, resistance to co-trimoxazole, streptomycin and neomycin varied during this period. Similar findings were also reported from cholera cases during 2001–2006 in Delhi; where resistance to ampicillin, furazolidone and nalidixic acid was observed (Ghosh *et al*., 2016). The changes in antibiotic susceptibility patterns shown by *V. cholerae* has been determined by horizontally transferable conjugative transposons found in the chromosome of the host bacterium (Waldor *et al*., 1996; Glass *et al*., 1980) or may be due to the presence of integrons (Mazel *et al*., 1998; Baker-Austin *et al*., 2009). As per previous reports, the *V. cholerae* O139 strains in India after 1997 showed a greater resistance profile towards ampicillin and neomycin and found susceptible to chloramphenicol and streptomycin (Mukhopadhyay *et al*., 1998). But the *V. cholerae* O139 emerged from 1999 to 2017 in Odisha were sensitive towards tetracycline and resistant to trimethoprim-sulfamethoxazole (SXT) and streptomycin. Various antibiotic-resistant genes such as *SulII*, *dfr*A1, *Str*B and *SXT* were positive for most of the *V. cholerae* O139 isolates during this period with some variations in 1999, 2003 and 2017. The patterns of a rapid shift in resistance to antibiotics are due to the substantial mobility of genetic elements such as *SulII*, *dfr*A1, *Str*B and *SXT*. Similar characteristics were also seen in *V. cholerae* O1 serogroups (Mukhopadhyay *et al*., 1996; Faruque *et al*., 1998b). Yamasaki *et al*. (1997) have reported that a multiple antibiotic-resistant plasmid belonging to incompatibility group C has also been attributed drug resistance to *V. cholerae* O139 strains (Yamasaki *et al*., 1997).

Pathogenesis to *V. cholerae* O1/O139 serogroups has been conferred by various genes that formed the genome of the filamentous CTXφ prophage. The multiplex PCR assays confirmed the presence of pathogenic genes such as *ompW*, *ctxA*, *tcpA*, *zot*, *rfbO139*, *ompU*, *rtxC*, *toxR* and *ctxB*. The occurrence of species-specific *ompW* and *rfbO139* genes confirmed the genus and serogroup of *V. cholerae* O139 respectively. The presence of *rtxC*, i.e., one of the gene in RTX toxin gene cluster encodes an acyltransferase, which is physically associated with the core element in the *V. cholerae* genome (Lin *et al*., 1999). In this study, various virulent genes like *toxR*, *zot*, *opmU* and *tcpA* were found positive in all the strains of *V. cholerae* O139 with some variations, suggesting the presence of core toxin region in O139 vibrios.

This study showed interesting patterns of genetic changes in CT gene present in CTXφ with varied allelic combinations of *rst*R gene (*rst*R^{Cl}, *rst*R^{ET} and *rst*R^{Calc}). The *V. cholerae* O139 isolates in 1999 possessed genotype 3, i.e., similar to the prototype El Tor strains but harboured a novel *rst*R allele, *rst*R^{Calc} type along with *rst*R^{ET}. The CT genotype 4 first appeared in Odisha in 2000 carried both types of *rst*R alleles, *rst*R^{Cl} and *rst*R^{Calc} (Table 3). The occurrence of Calcutta type CTX prophage (CTX$^{Calc}\varphi$) during 1996 in Kolkata differed from the earlier CTX$^{ET}\varphi$ phage primarily in the *rst*R gene that codes for the repressor protein of CTXφ (Kimsey and Waldor, 1998; Davis *et al*., 1999; Faruque *et al*., 2003). The genotype 4 of *V. cholerae* O139 is a variant of CT genotype 1 that exhibits change in single nucleotide only (cytosine instead of adenine) at position 83 (Raychoudhuri *et al*., 2010). It might have appeared due to a single mutation at CT genotype 1 that subsequently acquired by the re-emerged strains of *V. cholerae* O139 during 2000 cholera outbreak in Odisha. One *V. cholerae* O139 isolate in 2000 carried only one CTXφ genotype 1 with *rst*R^{Cl} allele. *V. cholerae* O139 isolates from 2002 onwards exhibited more than one copy of CTXφ and had *rst*R^{Cl}, *rst*R^{ET} and *rst*R^{Calc} alleles (Fig. 1). Similar results from Kolkata were published earlier, where *V. cholerae* O139 isolates harboured more than one copy of CTX prophage isolated from 2002, 2003 and 2005 respectively (Chatterjee *et al*., 2007; Raychoudhuri *et al*., 2010). The resurgence of O139 strains during 2017 after a hiatus of 10 years yielded amplicons for El Tor *ctx*B but possessed more than one copies of *rst*R allelels, *rst*R^{Cl}, *rst*R^{ET} and *rst*R^{Calc} in 42.8% strains (Table 2). Similar finding was reported from Delhi, India where out of 42 *V. cholerae* O139 isolates, three isolates in 2001 carried all the three *rst*R alleles (Ghosh *et al*., 2016). But a recent study reported the El Tor *ctx*B of *V. cholerae* O139 from 2017 had a point mutation at position 132 of amino acid sequence, where cysteine substituted by glutamine, being a new variant (Pal *et al*., 2019). Thus, the above findings revealed the presence of different allelic arrangements of *ctx*B and *rst*R due to the amalgamation of diverse CTX prophages among *V. cholerae* O139 isolates from Odisha. This study also confirmed that the genome of *V. cholerae* O139 is dynamic and had undergone cryptic alterations over the years since its emergence in 1992 (Fig. 1).

Additional information was obtained from *tcp*A gene analysis of the *V. cholerae* O139 strains that showed the dominance of El Tor *tcp*A allele (62.7%) over the years from 1999 to 2017. The toxin co-regulated pilus (TCP) is a homopolymer of the major pilus protein TCP pillin which encoded by *tcp*A that serves as a receptor for CTXφ. The classical and El Tor biotypes of *V. cholerae* differ slightly on the basis of C-terminal domain of the *tcp*A gene, whereas a alteration in amino acid sequence at position 64 of *tcp*A gene yields amplicons for Haitian variant *tcp*A. The presence of Haitian variant *tcp*A gene in *V. cholerae* O1 strains noticed in 1999 from Odisha was the first report from this region, India and globe. From the present study, the occurrence of Haitian variant *tcp*A gene in *V. cholerae* O139 strains was confirmed during 2000 and subsequently in 2006 from Odisha is a novel finding reported from this region, India and globe. So, this might be due to a horizontal gene transfer of Haitian variant *tcp*A gene from *V. cholerae* O1 to O139 strains through CTX prophages.

Table 3. Different combinations of *ctx*B and *rst*R of *Vibrio cholerae* O139 strains isolated from Odisha (1999–2017) after sequencing.

	Nucleotide position			Amino acid position				
Year	83	115	203	28	39	68	CT genotype	*rst*R type
1999	A	T	T	Asp	Tyr	Ile	Genotype-3	El Tor + Calcutta
2000	A	C	C	Asp	His	Thr	Genotype-1	Classical
2000	C	C	C	Ala	His	Thr	Genotype-4	Classical + Calcutta
2000	C	C	C	Ala	His	Thr	Genotype-4	El Tor + Calcutta
2000	A	T	T	Asp	Tyr	Ile	Genotype-3	El Tor + Calcutta
2002	C	C	C	Ala	His	Thr	Genotype-4	El Tor + Calcutta
2003	A	T	T	Asp	Tyr	Ile	Genotype-3	El Tor + Calcutta
2006	C	C	C	Ala	His	Thr	Genotype-4	El Tor + Calcutta
2007	C	C	C	Ala	His	Thr	Genotype-4	El Tor + Calcutta
2017	A	T	T	Asp	Tyr	Ile	Genotype-3	Classical + Calcutta

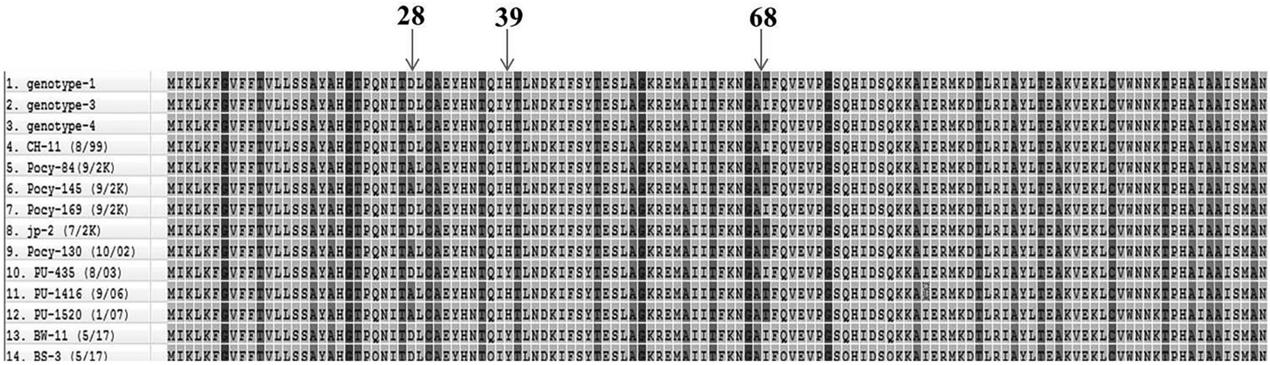
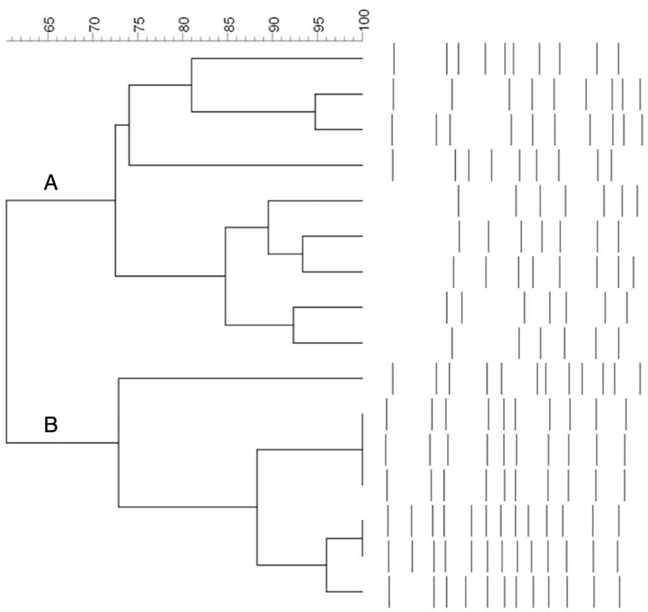


Fig. 2. Multiple sequence alignments of *ctx*B gene of corresponding *Vibrio cholerae* O139 strains from 1999 to 2017 showing changes in amino acid sequences at positions 28, 39 and 68.



Sample ID	Source	Isolation: Month & Year	Area
PU-1417	Rectal swab	09/2006	Coastal
JP-2	Rectal swab	07/2000	Tribal
POCY-198	Rectal swab	11/2000	Coastal
CH-11	Rectal swab	08/1999	Coastal
PU-130	Rectal swab	10/2002	Coastal
POCY-152	Rectal swab	09/2000	Coastal
POCY-43	Rectal swab	08/2000	Coastal
PU-1520	Rectal swab	01/2007	Coastal
PU-435	Rectal swab	08/2003	Coastal
KYW-59	Water	08/2017	Tribal
BW-11	Water	05/2017	Coastal
BS-20	Rectal swab	05/2017	Coastal
BS-19	Rectal swab	05/2017	Coastal
BS-9	Rectal swab	05/2017	Coastal
BS-5	Rectal swab	05/2017	Coastal
BS-3	Rectal swab	05/2017	Coastal

Fig. 3. PFGE patterns of *Not*I-digested *Vibrio cholerae* O139 strains with dendrogram analysis using Bionumeric 7.2 (Applied Maths, Belgium) showing 13 different pulsotypes arranged in two clusters, A and B.

The PFGE patterns of the *V. cholerae* O139 isolates from 1999 to 2017 showed diversity of their genomes containing 13 different pulsotypes arranged in two clusters. The *V. cholerae* O139 isolates of 2017 were different from the strains which were isolated during 1999–2007 with 60% similarity. The difference in pulsotypes of 2017 *V. cholerae* O139 isolates in comparison to other years might be due to the new variant of El Tor *ctx*B of *V. cholerae* O139 isolates (Pal *et al*., 2019). Earlier studies demonstrated the genomic diversities of *V. cholerae* O139 isolates from Delhi, India (2001–2006) and also from Guangdon, China (1993–2013) that clearly showed the relatively high clonal expansions within the *V. cholerae* O139 isolates (Ghosh *et al*., 2016; Li *et al*., 2016). The present findings on PFGE analyses clearly evident that the genomes of *V. cholerae* O139 has undergone several adaptations and evolution since its first emergence in 1992 and spread to Odisha from 1999 to 2017.

Conclusion

The above study concluded that *V. cholerae* O139 strains isolated from 1999 to 2017 in Odisha are showing genotypic variations evident from *ctx*B gene sequencing and *rst*R allelic combinations. The presence of Haitian variant *tcp*A gene in *V. cholerae* O139 strains of 2000 and 2006 is a unique and new report from Odisha and the globe. Although multiple clonal variations are seen between *V. cholerae* O139 strains isolated over the years from coastal and tribal areas, more genomic studies like multilocus-sequence typing (MLST) and whole genome sequencing are warranted to study the genomic diversities of *V. cholerae* O139 strains from this region.

Experimental procedures

Bacterial strains

Fifty-nine isolates of *V. cholerae* O139 were revived from the laboratory stocks collected previously from the diarrhoea patients and environmental water samples during diarrhoea surveillance studies, cholera outbreaks and epidemics from 1999 to 2017 from different districts of Odisha. All the strains were cultured on thiosulphate citrate bile salt sucrose agar (TCBS) (BD, USA) and incubated at 37°C for 16–18 h. The developed yellow colonies were further sub-cultured on Luria agar (BD) and then subjected to different biochemical identification test (Nair *et al*., 1987). These *V. cholerae* O139 isolates were finally confirmed serologically using *V. cholerae* O139-specific antiserum.

Antibiotic susceptibility test

All the *V. cholerae* O139 strains were analysed for their antibiotic susceptibility by disk diffusion method and as per Clinical and Laboratory Standard Institute (CLSI) guidelines using standard procedures (Bauer *et al*., 1966; Wayne, 2010). The antibiotics impregnated disks (Difco, USA) used were ampicillin (Am, 10 μg), chloramphenicol (C, 30 μg), co-trimoxazole (Co, 25 μg), ciprofloxacin (Cf, 5 μg), furazolidone (Fr, 100 μg), gentamicin (G, 10 μg), neomycin (N, 30 μg), nalidixic acid (Na, 30 μg), norfloxacin (Nx, 10 μg), streptomycin (S, 10 μg) and tetracycline (T, 30 μg).

PCR analysis

Preparation of DNA template. Bacterial genomic DNA was extracted by the boiling method from overnight grown culture in Luria-Bertani (LB) broth at 37°C. About 1 ml aliquot of the LB broth was taken into a sterile 1.5 ml micro centrifuge tube and then centrifuged at 50 000*g* for 5 min. The cell pellet thus obtained was boiled for 15 min in a water bath and then immediately transferred to −20°C for further use (Pal *et al*., 2017).

Analysis of virulence genes. The *V. cholerae* O139 strains were analysed through two separate multiplex PCR (mPCR) assays for their serogroups, toxigenicity and pathogenicity traits following the methods described elsewhere (Kumar *et al*., 2009). The mPCR-I detected various genes, such as *ompU*, *ctxB*, *toxR*, *rfbO139* and *rtxC*. The mPCR-II was used to confirm the presence of virulence and toxigenic genes such as *OmpW*, *ctxA*, *tcpA* and *zot*. The PCR assays were performed containing 25 μl reaction mixture as follows: pre-incubation at 94°C for 2 min; denaturation at 94°C of 1 min (30 cycles), annealing at 58°C for 1 min and then primer extension at 72°C for 2 min followed by final extension at 72°C for 10 min.

Analysis of antibiotic resistance genes. The *V. cholerae* O139 strains were analysed for the presence of antibiotic-resistance genes that encodes for sulfamethoxazole (*sulII*), trimethoprim (*dfrA1*), streptomycin (*strB*) and SXT genetic element using multiplex PCR assay. Multiplex PCR assay was carried out as follows: pre-incubation at 94°C for 2 min, denaturation at 94°C for 1 min (35 cycles), annealing at 60°C for 1 min and primer extension at 72°C for 1 min followed by final extension at 72°C for 10 min. The PCR products were run in 1.8% agarose gel, stained with ethidium bromide and visualized under a gel documentation system (Ramachandran *et al*., 2007).

PCR for biotyping. The *ctx*B gene analysis of 59 *V. cholerae* O139 strains isolated from 1999 to 2017 was done using MAMA PCR that differentiate between genotype 1 (classical type CT) and genotype 3 (El Tor CT) based on the nucleotide position 203 of the *ctx*B gene (Morita *et al*., 2008). Molecular characterization for the identification of variants of *tcp*A gene (Classical, El Tor and Haitian) was done using three specific primers (One common EL-Rev primer and two forward primers F1/F2 for Haitian and El Tor-specific primers, respectively) (Ghosh *et al*., 2014). The types of *rst*R alleles present in the *V. cholerae* O139 strains were identified by PCR assay using three different forward primers specific for *rst*R^{Cl}, *rst*R^{ET} and *rst*R^{Calc} with a common reverse primer *rstA3*R (Kimsey and Waldor, 1998; Nusrin *et al*., 2004). Detailed list of primers used in this study is given in the Supporting Information Table S1.

Nucleotide sequencing of ctxB gene. The nucleotide sequence analyses of the *ctxB* genes were performed after PCR amplification of *ctxB* locus of 10 representative strains of *V. cholerae* O139. PCR assay was carried out with PCR primers and conditions as described previously (Mitra *et al*., 2001). The gel purification kit (GCC Biotech, India) was used for the purification of PCR products and then sequencing was done in an automated DNA sequencer (ABI PRISM® 377, Applied Biosystems). The nucleotide sequences thus obtained were compared with the sequences available in the GenBank by multiple sequence alignment using CLUSTAL W in MEGA 7.0. The nucleotide sequence data of *V. cholerae* O139 strains were submitted to the GenBank with accession numbers: MW067005, MW067006, MW067007, MW067008, MW067009, MW025955, MW025956, MW046333, MW046334, and MW046336.

Pulsed-field gel electrophoresis. The *NotI*-digested genomic DNA of representative strains of *V. cholerae* O139 from each year was subjected to PFGE analysis following the PulseNet standardized PFGE protocol using CHEF Mapper System (Bio-Rad) (Cooper *et al*., 2006).

Acknowledgements

The authors would like to thank the Director, ICMR-Regional Medical Research Centre, Bhubaneswar, Odisha for providing intramural assistance from the institute to conduct this study.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Table S1. PCR primers used for the study of *Vibrio cholerae* O139 genes.